Mingfeng He

Shanghai Jiao Tong University, P. R. China

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EDUCATION

School of Mathematical Sciences, Zhiyuan Honors Program (For top 10% students), Shanghai Jiao Tong University, Shanghai, China Aug 2022 - present

Major GPA: 3.78

Selected Core Courses:

Mathematical AnalysisII (92), Mathematical AnalysisIII (93), Professional SeminarI (95), Professional SeminarII (91)
Ordinary Differential Equation (94), Introduction to Physics (95), Abstract Algebra (97), Differential geometry (98)

School of Engineering Sciences, KTH Royal Institute of Technology, Exchange Program in Mathematics, Stockholm, Sweden Jan 2025 – June 2025

Overall GPA:4.0/4.0

Selected Courses: Applied linear optimization, Applied nonlinear optimization.

PUBLICATIONS

- Yu Cao, **Mingfeng He**, Xiantao Li. **Dynamically Optimal Unraveling Schemes for Simulating Lindblad Equations.** arXiv: 2509.19887 (preprint)
- Yangfei Liao, Zhongshuo Lin, Jianghao Liu, Qingyuan Sun, Yifan Wang, Teng Wu, Hehu Xie, **Mingfeng He. Solving Schrodinger Equation Using Tensor Neural Network.** arXiv: 2209.12572 (preprint)

RESEARCH EXPERIENCES

The research on the extension of Schwarz Lemma | Shanghai Jiao Tong University Oct 2023 – Nov 2024

Advisor: Feng Rong, Professor at School of Mathematical Sciences, Shanghai Jiao Tong University

- Reviewed Rudin's *Function Theory on the Unit Ball* for a foundational understanding of the Lemma in multiple complex variables, alongside Kim Lee's *Schwarz's Lemma from a Differential Geometric Viewpoint* for its geometric interpretation.
- Completed a project report of over 60 pages. The main content of the report includes a summary and overview of three research directions: geometry, boundary, and high-dimensional research.

Tensor neural network(TNN) for solving partial differential equations | Academy of Mathematics and Systems Science, Chinese Academy of Sciences (AMSS, CAS) Jul 2024 – Sept 2024

Advisor: Hehu Xie, Professor at the Academy of Mathematics and Systems Science, Chinese Academy of Sciences.

- Explored the fundamentals of using TNN to solve partial differential equations and successfully reproduced the code from the original paper "Solving Schrödinger Equation Using Tensor Neural Network".
- Conducted further research on TNN interpolation and proposed a new method to calculate Coulomb potential, significantly improving both accuracy and computational efficiency. Applied TNN and a novel interpolation method to calculate the ground-state energy of helium, lithium, and hydrogen, achieving accuracy improvements of up to four orders of magnitude.

Optimal Unraveling Schemes for Simulating Lindblad Equations. | Shanghai Jiao Tong University Sept 2024 - Sept 2025

Advisor: Yu Cao, Associate professor at School of Mathematical Sciences, Shanghai Jiao Tong University.

- Developed a complete parametric characterization of all diffusion- and jump-type stochastic unraveling schemes for Lindblad equations.
- Derived an explicit dynamically optimal quantum state diffusion (DO-QSD) scheme by minimizing the short-time growth of observable variance.
- Demonstrated through numerical experiments that DO-QSD reduces trajectory variance by roughly one order of magnitude, outperforming both conventional QSD and machine-learning–trained schemes.

Kolmogorov–Arnold Network Enhanced PINNs with Self-Scaled BFGS Optimizer for Inverse PDE Problems. | The Chinese University of Hong Kong Jul 2025 - Aug 2025

Advisor: Bangti Jin, Professor at the Department of Mathematics, The Chinese University of Hong Kong (CUHK).

- Explored combining Kolmogorov–Arnold Networks (KAN) with Physics-Informed Neural Networks (PINNs) for inverse PDE problems, showing about one-order-of-magnitude lower relative error under the same conditions.
- Evaluated the self-scaled BFGS optimizer proposed by Thomas Y. Hou et al., and verified its high performance in PDE inverse problems.

Tensorized Operator Learning with Kolmogorov–Arnold Networks for Uncertainty Quantification. | Purdue University Apr 2025 – present

Advisor: Guang Lin, Associate Dean for Research & Innovation, College of Science; Director, Data Science Consulting Services; Full Professor, Departments of Mathematics, Statistics, and School of Mechanical Engineering, Purdue University

- Developed tensorized extensions of the Deep Operator Network (Tensor-DeepONet) to efficiently represent high-dimensional operators and quantify predictive uncertainty. Introduced Kolmogorov–Arnold Networks (KANs) to enhance local adaptability and interpretability within operator-learning frameworks. Incorporated conformalized prediction techniques to achieve calibrated uncertainty quantification for complex-valued outputs.
- Applied the proposed model to the two-dimensional Schrödinger equation, achieving a 50– 60% lower reduction in test error compared with standard DeepONet and FNO baselines while accurately reconstructing amplitude and phase fields.
- Currently extending the framework to inverse problems and higher-dimensional PDEs, focusing on efficient operator representation and reliable uncertainty estimation; a manuscript is in preparation.

COURSE PROJECTS

- Continuous but nowhere differentiable functions including fractal theory — Mathematical AnalysisII
- Optimization models and numerical methods for Biological Transport Network — Professional SeminarII
- Encoding capabilities of E-I Balanced Networks — Professional SeminarIII

AWARDS

- Zhiyuan Honor's Scholarship — December 2022,2023,2024
  Reward outstanding students from Zhiyuan College in 2022,2023,2024
- The C-level Scholarship for Undergraduates in 2023,2024 — December 2023,2024
  School level scholarships, ranking from 15% to 30%

SKILLS

Language Skills: English (TOEFL iBT 104), Chinese (native)

Programming Skills: Python, Matlab, GAMS, Latex

GRE Mathematics Subject Test: 900 (91st percentile)